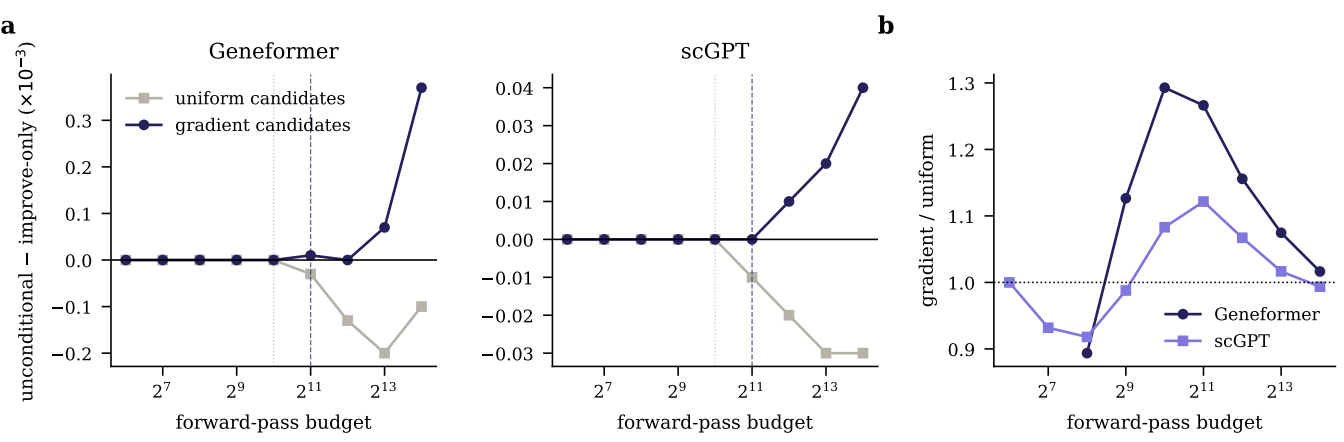
a
Geneformer
scGPT
uniform candidates
gradient candidates
unconditional − improve-only (×10⁻³)
forward-pass budget
forward-pass budget
b
gradient / uniform
Geneformer
scGPT
forward-pass budget